

Knowledge Graph Traversal and Entity Extraction

Graph RAG Concepts

Knowledge Graph RAG supplements dense embeddings by capturing explicit entity relationships (subject-predicate-object triples). Graph structures maintain structural integrity across multi-hop inference paths where semantic vector similarity falls short.

Entity & Relation Extraction

NER models and LLM extractors scan unstructured text to identify core entities and directed edges. Extracted knowledge triples are stored in graph databases like Neo4j or NetworkX for relational query execution.

Sub-graph Retrieval

During retrieval, seed entities identified in the user prompt are expanded via personalized PageRank or breadth-first search (BFS) to construct a context sub-graph.